

```
1 clc;
2 clear;
3 close all;
4
5 snr = 0:5:30;
6 snr_linear = 10.^(snr/10);
7 capacity = log2(1 + snr_linear);
8
9 antennas = [4 8 16 32 64];
10 gain = 10*log10(antennas);
11
12 figure;
13
14 subplot(3,1,1);
15 plot(snr, capacity, 'o-', 'LineWidth', 2, 'MarkerSize', 6);
16 grid on;
17 xlabel('SNR (dB)');
18 ylabel('Channel Capacity (bps/Hz)');
19 title('Channel Capacity vs SNR');
20
21 subplot(3,1,2);
22 plot(snr, capacity, 'o-', 'LineWidth', 2, 'MarkerSize', 6);
23 grid on;
24 xlabel('SNR (dB)');
25 ylabel('Channel Capacity (bps/Hz)');
26 title('Channel Capacity vs SNR');
27
28 subplot(3,1,3);
29 bar(antennas, gain);
30 grid on;
31 xlabel('Number of Antenna Elements');
32 ylabel('Antenna Gain (dB)');
33 title('Antenna Gain for Different Antenna Array Sizes');
```

